

Chapter 3: Choice under Uncertainty

1 Introduction

Much of the discussion in economics so far has assumed that consumers make decisions with perfect knowledge on the outcomes. In reality, the choices we make are often done under uncertainty. It is important for economic analysis to incorporate how agents decide between decisions when the outcomes are probabilistic.

2 Expected Utility Theory

Let X represent a random variable that can take values $x_1, \dots, x_n$ with probability $p_1, \dots, p_n$. The expected value of X is given by:

$$E(X) = p_1x_1 + \dots + p_nx_n = \sum_i p_ix_i$$

In statistical analysis, we often denote X as some kind of outcome, such as a dice roll, a card drawn from a deck, profits from an asset, or the yield from a farm. In microeconomic theory, we simply extend the analysis where the values X can be transformed into a utility function $u(X)$:

$$E[U(X)] = p_1u(x_1) + \dots + p_nu(x_n)$$

For example, consider a farmer that plants seeds on their land. The weather is good with probability $p_1 = 0.8$ which yields the farmer $x_1 = 100$. If the weather is bad, then the value of the yield is $x_2 = 10$.

The expected yield from this farm is derived as:

$$E(X) = 0.8(100) + 0.2(10) = 82$$

However, let us say the farmer has a utility function which they derive from this yield:

$$U(X) = \sqrt{X}$$

We can say that the expected utility from the outcome is:

$$E[U(X)] = 0.8 \cdot \sqrt{100} + 0.2 \cdot \sqrt{10} = 8.63$$

3 Risk Attitudes

When utility is represented by expectations, it is called a von Neumann-Morgenstern (vN-M) utility function^a. We are going to use this to represent a person's tolerance towards making decisions when the outcomes are uncertain.

In this chapter, we explore three different scenarios of an individual which is *risk averse*, *risk-neutral* and *risk-seeking*.

To fix ideas, consider an individual who starts with an income of 500 BD. They have a choice to invest it in an asset or to do nothing and keep the money. The asset has a 50% chance of increasing in value to 750 BD and 50% chance to decrease to 250 BD.

Something to keep in mind is that the expected value of the asset is the same as doing nothing:

$$E(X) = 0.5(750) + 0.5(250) = 500.$$

This is called a *fair gamble*, and we will explore how individuals with different risk attitudes will engage with this decision.

3.1 Risk Averse

An individual is considered risk averse if they have a concave utility function. Formally, we would say that:

$$U'(X) > 0 \quad U''(X) < 0,$$

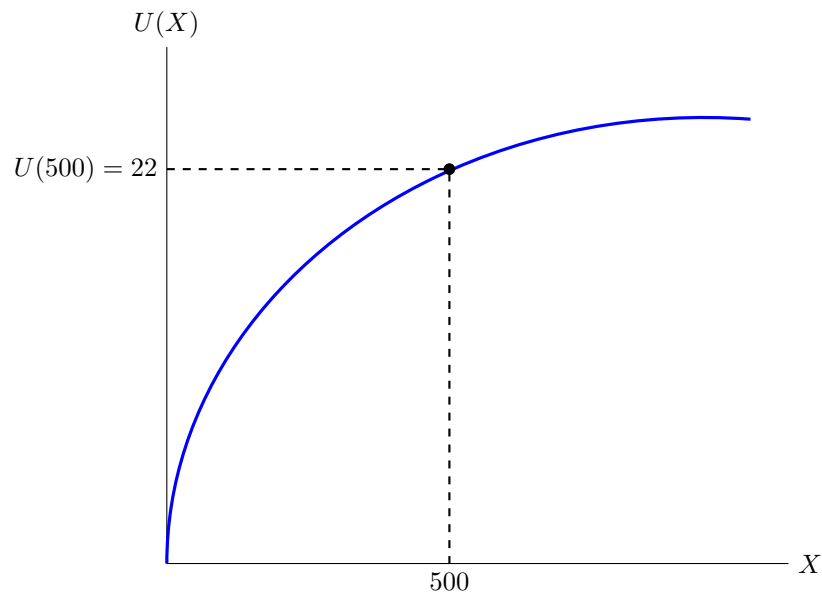
which means that utility increases at a decreasing rate with respect to the outcome.

Let us say that the individual deciding on the asset is risk averse and has the following utility function:

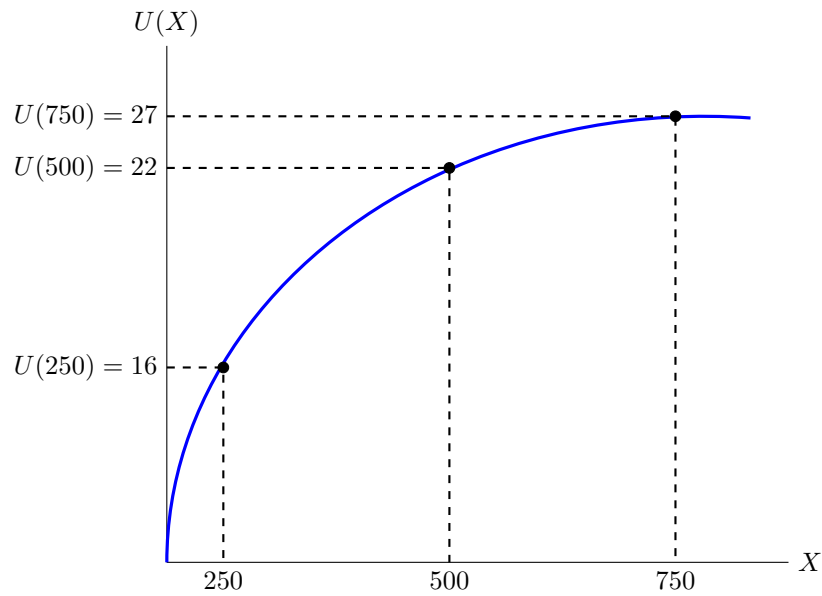
$$U(X) = \sqrt{X}$$

We can illustrate the initial position of doing nothing in the below plot:

^aThe regular utility function we learnt in the previous chapter allows all forms of monotonic transformations (squaring, logging, etc.). On the other hand, the vN-M utility function only allows positive affine transformations (such as $a + bx$). This is because any other change would disrupt the order of rankings between choices. See Problem Set 3 for a detailed example.



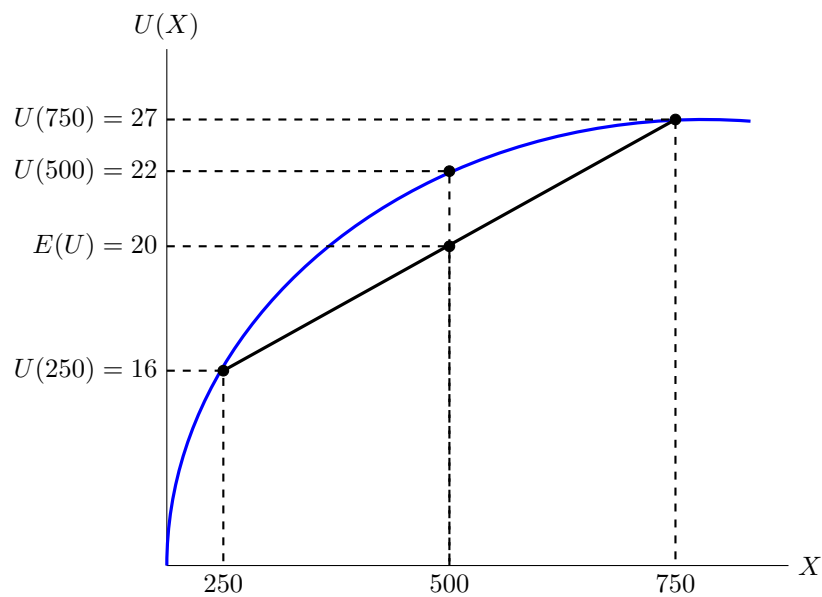
Now let us calculate the individual's utility if they decided to invest in the asset. The good and bad outcome can be shown as:



Now let us calculate the expected utility of investing in this asset:

$$E(U(X)) = 0.5\sqrt{250} + 0.5\sqrt{750}.$$

Geographically, this is actually the midpoint between the good and bad outcomes:



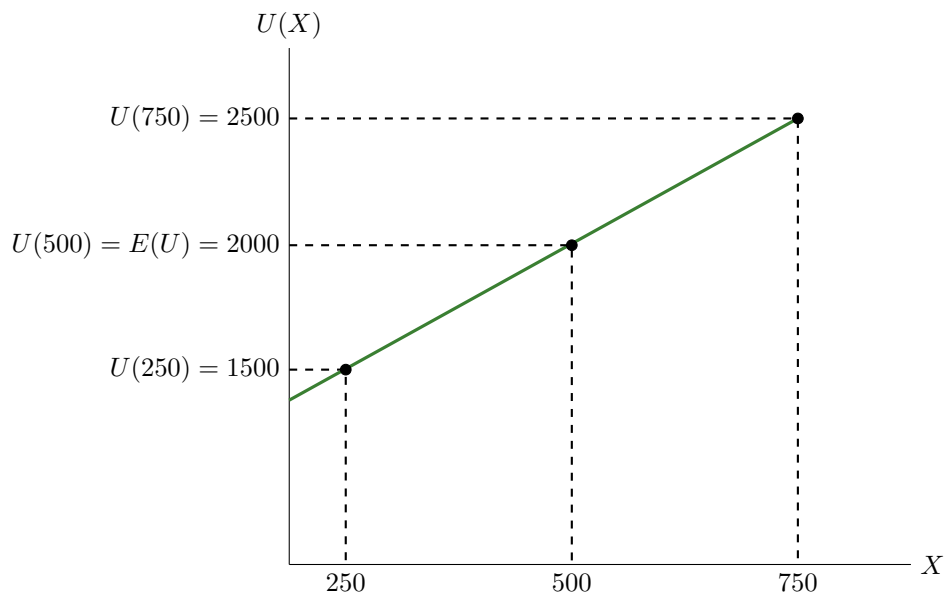
We can see that the utility gained from not engaging in this investment yields the individual $U(500) = 22$. However, given that the asset yields an uncertain outcome, the expected utility from investing in it is $E(U) = 20$. Therefore, a risk averse individual would reject a fair gamble.

3.2 Risk Neutral

A risk neutral individual can be formally characterised as having a linear utility function, such as:

$$U(X) = 1000 + 2X$$

Let us repeat the analysis of calculating the utility of doing nothing compared to investing in the asset:



We can see that in this case, a risk neutral individual is indifferent between doing nothing and investing in the asset, since the fair game yields the same utility.

3.3 Risk Seeking

Formally, an individual is considered risk-seeking if they have an exponential utility function:

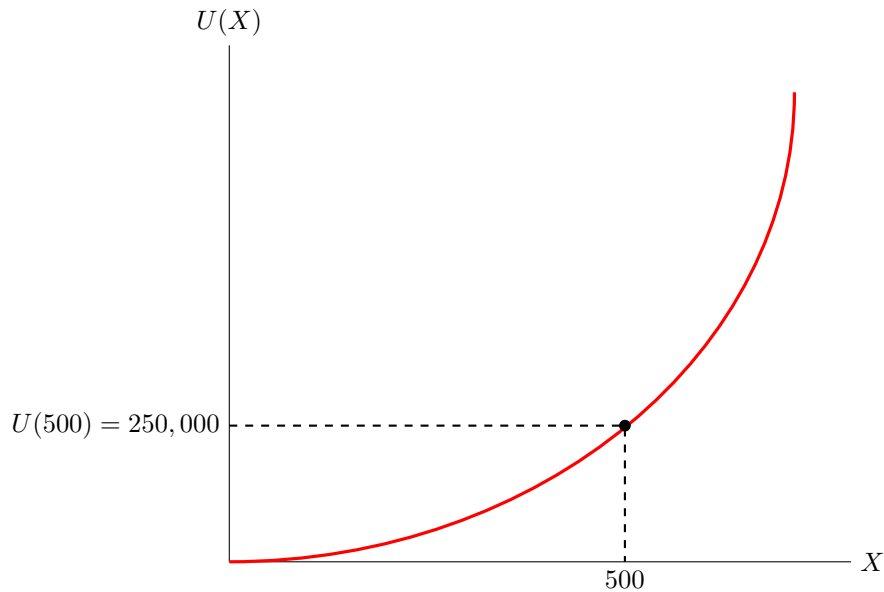
$$U'(X) > 0 \quad U''(X) > 0,$$

which means that utility increases at an increasing rate with respect to the outcome.

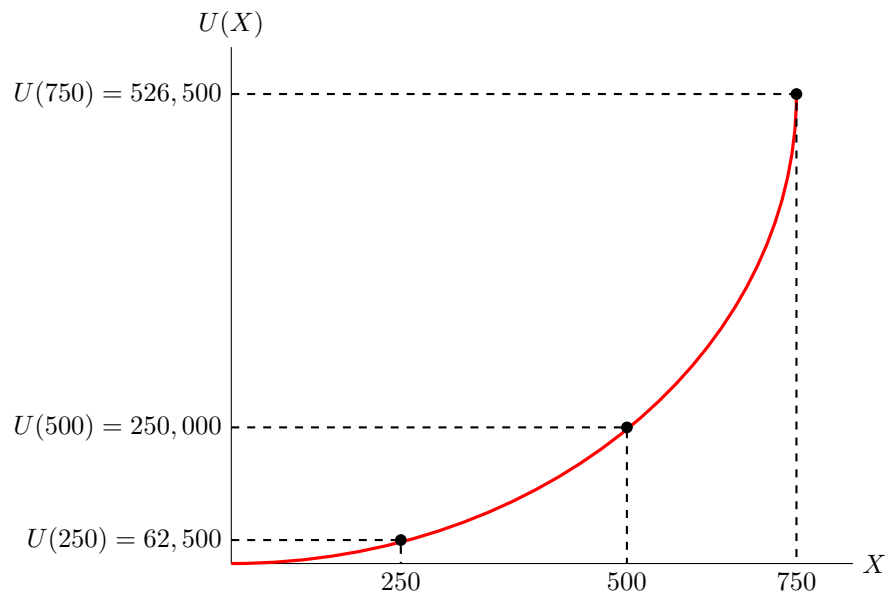
For example, consider the individual having a risk-seeking utility function:

$$U(X) = X^2$$

We can illustrate the initial position of doing nothing:



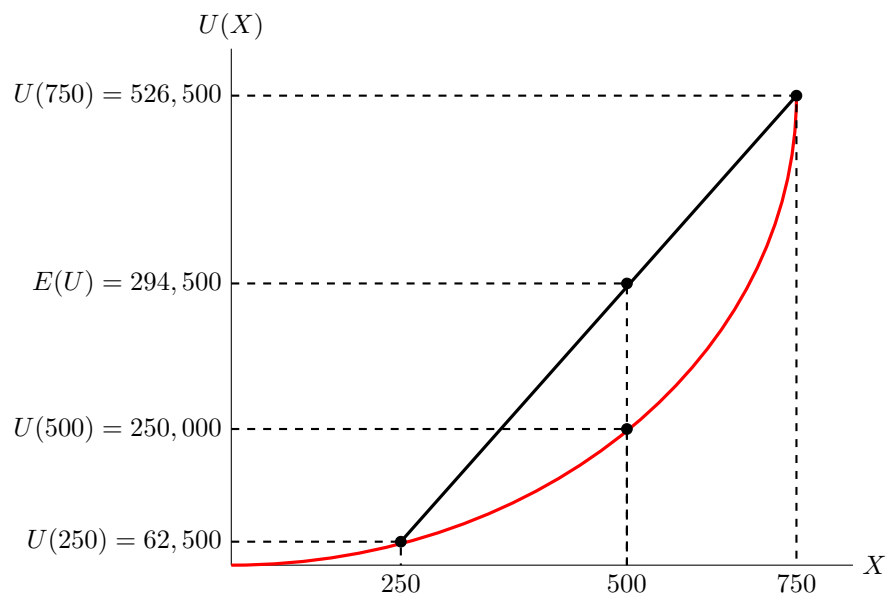
Now let us calculate the individual's utility if they decided to invest in the asset. The good and bad outcomes can be shown as:



Now let us calculate the expected utility of investing in this asset:

$$E(U(X)) = 0.5(250)^2 + 0.5(750)^2$$

Similar to the first scenario, this is geographically the midpoint between the good and bad outcomes:



We can clearly see that the expected utility gained from investing in the asset $E(U) = 294,500$ is greater than the utility of doing nothing $U(500) = 250,000$. This means that a risk-seeking individual would prefer investing in this fair gamble.

4 Certainty Equivalent

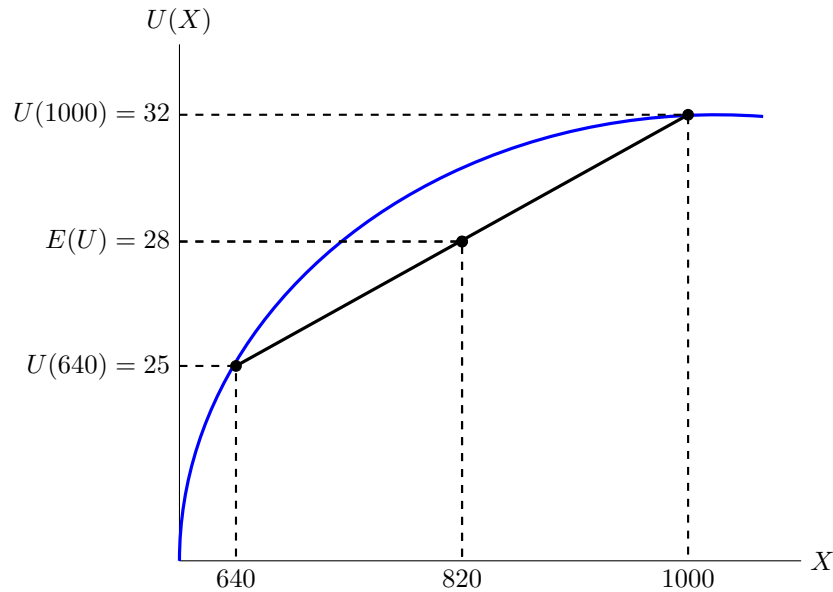
Let us now evaluate how an individual behaves if they have no choice but to engage in a decision with uncertain outcomes. In this section, we will explore the idea of *insurance* and how it is used to mitigate the risks they face. We will see that the willingness to spend on insurance will depend on their risk attitude.

Suppose an individual earns a monthly wage of 1000 BD and faces uncertainty with regards to their health outcome. Every month, there is a 50% chance that nothing happens, but there is also a 50% that they become sick and will spend 360 BD on medical expenses.

Let us examine the case of a risk-averse individual who has a concave utility function:

$$U(X) = \sqrt{X}$$

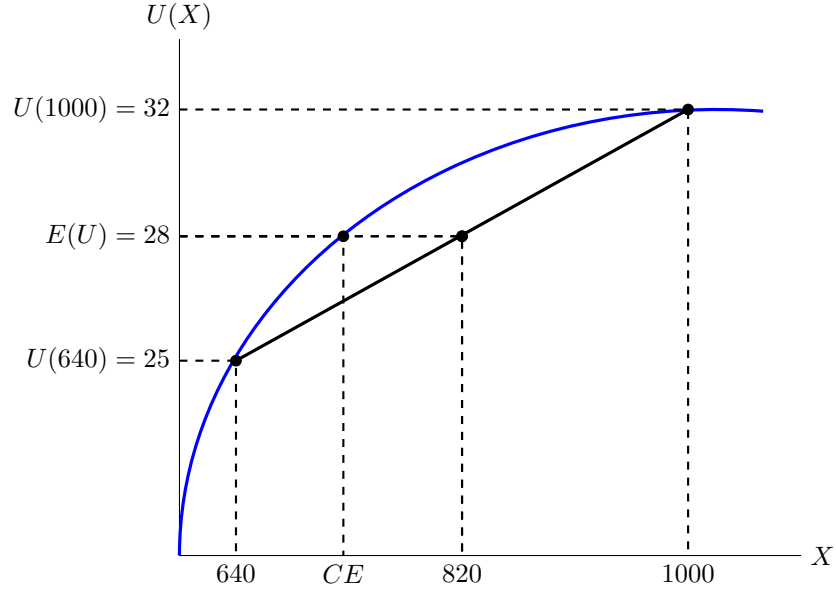
Unlike the previous case, the individual has no choice but to face this uncertainty (they cannot opt out of the gamble). We can illustrate the uncertainty in the below plot:



Now consider if the individual has an option to purchase insurance. If they agree to pay a premium charge R per month, then the insurance company will pay for all medical expenses if the adverse event occurs^b. We are interested in how much the individual is willing to pay per month in order to make it worthwhile to purchase this insurance policy.

Intuitively, the maximum an individual is willing to pay is an amount R that **makes them indifferent** on their decision to purchase insurance. Mathematically, this is when the utility outcome is identical:

^bTo be clear, if the individual starts with 1000 BD, but has to pay medical expenses of 360 BD in this period, then the insurance company agrees to pay this amount immediately. However, the individual still needs to pay the premium R . This means that the net income is $1000 - R$.



The Certainty Equivalent CE is the amount of money that makes the individual indifferent between having and not having insurance. We can use this fact to derive the maximum^c that an individual is willing to pay in terms of insurance premiums.

Let us first derive the CE :

$$U(CE) = E(U)$$

$$\sqrt{CE} = 28$$

$$CE = 810.$$

When the adverse event occurs, the individual is compensated for their loss, which means their income is 1000 BD again. However, they still need to pay the premium even if the loss occurs. Therefore, the maximum R that an individual will consider paying is when their income is at the CE level:

$$R = 1000 - 810 = 190.$$

Therefore, a risk averse individual will pay a maximum of $R_{\max} = 190$ per month.

In some cases, it is also interesting to know what the minimum level of insurance would be offered to the individual. In this case, we look at the perspective of the insurance company.

Let us assume the insurance company always offers a premium that is *fair*. This means that they charge a rate that ensures they break even in terms of their costs. If they are expected to pay 360 BD with 50% probability in every period, then in order to break even across periods, the premium must at least be $R_{\min} = 0.5(360) = 180$ BD. This describes the minimum premium that would be offered, and if we forced the insurer to be fair, then it would always charge this rate instead of $R_{\max}$.

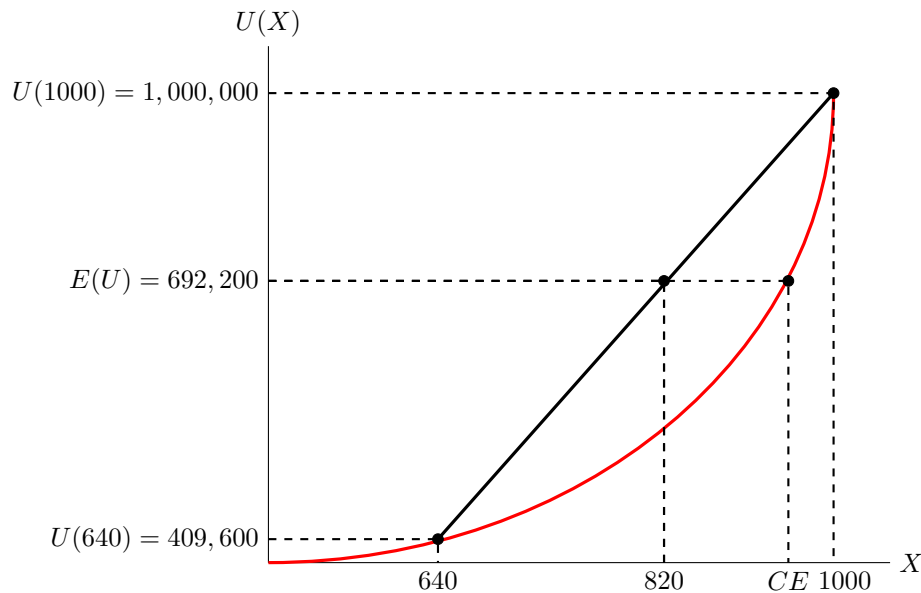
^cIf the individual does not have insurance, their expected wealth per month is 820 BD. Let us say they agree to have insurance, but only pay 5 BD per month in premiums. This means that the expected wealth of the individual is 995. If it was 10 BD, then expected wealth is 990. If it was 50 BD, it would be 950. We are interested in finding out the maximum R that the individual will consider satisfactory to purchase insurance. This occurs when the utility from their wealth (income minus premium) is identical to their utility on wealth without insurance $E(U) = 28$. The level of wealth where these are identical is called the Certainty Equivalent.

Example 3.A: Consider an identical scenario as discussed above, but that this time, the individual is risk-seeking with a vN-M utility function:

$$U(X) = X^2.$$

Illustrate the scenario graphically and provide an interpretation for the CE in this case.

Solution: The difference now is that the individual has a utility function with an exponential shape:



The CE can be calculated as:

$$U(CE) = E(U)$$

$$CE^2 = 692,200$$

$$CE = 832$$

This is the amount of wealth the individual would need to have in order to make them indifferent between having and not having insurance.

We can see that this is higher than their expected wealth under no insurance. We can interpret this as the individual would need to be *paid* a premium amount in order to make them indifferent.

5 Choice of Insurance

The above examples highlighted the concept of *full* insurance. This means that the insurer would recover the complete loss suffered by the individual. In reality, insurance packages tend to have certain limits depending on how much

premium a person is willing to pay. We can formally derive a general condition that highlights this.

Suppose an individual who has a wealth W , but faces the prospect of loss L with probability p , where $0 < p < 1$. The individual can purchase a coverage X by paying the premium rX at every period^d. We are interested to see what is the optimal coverage value that the individual will purchase.

For the two outcomes, we can say:

- Bad Outcome: $W_L = W - L + X - rX$ with probability p .
- Good Outcome: $W_N = W - rX$ with probability $1 - p$.

For a general utility function, $U(W)$, we can derive the expected utility of the individual facing these outcomes:

$$\begin{aligned} E(U) &= pu(W - L + X - rX) + (1 - p)u(W - rX) \\ &= pu(W_L) + (1 - p)u(W_N). \end{aligned}$$

Remember that the individual is deciding on the optimal level of coverage. This means they maximise their objective function $E(U)$ with respect to X and set first order conditions^e:

$$\frac{\partial E(U)}{\partial X} = (1 - r)pu'(W_L) - r(1 - p)u'(W_N) = 0.$$

This gives us a general statement as to what level of coverage an individual will purchase from the insurance company.

Consider the health outcome example. We can prove that when an individual is risk-averse and that when the insurance premium is fair, then the individual will optimally purchase full insurance.

Let us first arrange the equation:

$$\frac{u'(W_L)}{u'(W_N)} = \frac{(1 - p)r}{(1 - r)p}.$$

Since the insurance company charges a fair rate, this means that they always break even. The expected payout to the individual pX is equal to the expected revenue from the individual rX :

$$pX = rX$$

$$p = r.$$

Substituting this in the above equation:

^dIn the context of the previous example, $X = 360$ since the individual is purchasing full insurance, to which they paid a rate 19 every period. In this example, we are making a general statement of what X can be and allowing the premium rate r to scale with the coverage.

^eYou need to use the chain rule.

$$\frac{u'(W_L)}{u'(W_N)} = \frac{(1-r)r}{(1-r)r}$$

$$\frac{u'(W_L)}{u'(W_N)} = 1$$

$$u'(W_L) = u'(W_N).$$

Since the individual is risk-averse, the utility function means that it is not possible for $W_L \neq W_N$. This is because $u'' < 0$, which means that there is only a single global maximum point^f. It follows that:

$$W - L + X - rX = W - rX$$

$$X = L,$$

which means that a risk averse individual that is being charged a fair insurance rate will purchase a coverage X that covers their entire losses L . We saw this exact result in the health outcome example.

Example 2.B: Suppose that the insurance company did not charge a fair rate, and instead sets:

$$rX > pX$$

$$r > p,$$

which means that it on average makes more money from the individual whenever the adverse outcome occurs. Show that a risk-averse individual will instead purchase a coverage that is less than their loss $X < L$ in this case.

Solution: Going back to the first order conditions:

$$\frac{u'(W_L)}{u'(W_N)} = \frac{(1-p)r}{(1-r)p} > 1,$$

since $r > p$, the right hand side is greater than 1. This must mean that:

$$u'(W_L) > u'(W_N).$$

Since the individual is risk-averse^a, then it must be the case that $W_L < W_N$:

^fIf this were not the case, then it would be possible for $u(\cdot)$ to have two or more peaks that give the same level of utility but at different converge points.

$$W - L + X - rX < W - rX$$

$$X < L.$$

This demonstrates that when insurance is not charged fairly, then a risk-averse individual simply purchases less than full insurance to compensate for this.

A profit maximising insurance company will always try to offer a premium that is higher than the probability of the adverse event occurring. Moreover, an individual does not usually know how much they will pay in medical expenditures over their life time. This asymmetric information leads to a form of market failure, and we will address this in later chapters.

^aConsider a concave function $y = f(x)$. Notice it starts off very steep and gradually turns flat as you increase x . Formally, $\frac{\Delta y}{\Delta x} \rightarrow 0$ as $x \rightarrow \infty$, which shows that the derivative gets smaller as x gets larger. Therefore, if the derivatives $u'(W_L) < u'(W_N)$, then it must be that the $W_L > W_N$.

6 Measuring Risk Attitudes

So far in our discussions, we have made general statements about risk attitudes, in that they are either averse, neutral or risk-seeking. Realistically, all individuals have varying levels of risk tolerance. For example, an individual can be *more* risk-averse relative to another risk-averse person. The degree of risk-attitudes implies that each individual behaves differently depending on the curvature of their vN-M utility function. This information is useful when deriving higher level models in microeconomic theory, but we only discuss the basic concept here.

One way to measure a person's risk attitude is with the **Arrow-Pratt** method:

$$\rho = -\frac{u''}{u'}$$

It is easy to see that depending on the risk attitude of an individual:

- $\rho > 0$ if they are risk-averse, since $u'' < 0$.
- $\rho = 0$ if they are risk-neutral, since $u'' = 0$.
- $\rho < 0$ if they are risk-seeking, since $u'' > 0$

Example 3.1: For each utility function, derive the Arrow-Pratt measure of risk attitude:

(a) $u(W) = \ln W$.

(a) $u(W) = W^a$ where $0 < a < 1$.

(a) $u(W) = -e^a$.

Solution Remember that the formula is:

$$\rho = -\frac{u''}{u'}$$

Therefore, we need to derive the first and second derivative of all the vN-M utility functions.

(a) For $u(W) = \ln W$, it follows that:

$$\begin{aligned}\rho &= -\frac{-\frac{1}{W^2}}{\frac{1}{W}} \\ &= \frac{1}{W^2} \cdot W = \frac{1}{W}\end{aligned}$$

This clearly shows that the degree of risk aversion is decreasing in wealth, so the individual is risk-averse.

(b) For $u(W) = W^a$, and that $0 < a < 1$, it follows that:

$$\begin{aligned}\rho &= -\frac{(a-1)aW^{a-2}}{aW^{a-1}} \\ &= -(a-1)W^{-1} \\ &= \frac{1-a}{W}\end{aligned}$$

This individual is also risk-averse, but note that as $a \rightarrow 1$, then $\frac{1-a}{W} \rightarrow 0$, which means the individual becomes risk-neutral.

(c) For $u(W) = -e^{-a}$, it follows that:

$$\begin{aligned}\rho &= -\frac{-a^2 e^{-a}}{a e^{-a}} \\ &= a\end{aligned}$$

In this case, the degree of risk aversion is independent of wealth.

7 Problem Set

1) Consider a risk averse individual that has to choose between two decisions:

- A gamble that pays 1000 BD with probability $1/4$ and 100 BD with probability $3/4$.
- Receive an instant payment of 325 BD.

(a) Determine which decision the individual will choose.

(b) Show that if the individual was offered 320 BD instead as a direct payment, then the answer will depend on the degree of risk aversion.

2) Suppose $u(W) = -\frac{1}{W}$. What is the risk attitude of this person? Calculate the Arrow–Pratt measure of risk aversion for this preference.

3) Suppose an agent has vN–M utility function $u(W)$. Under what condition would $a + bu(W)$ also be a valid vN–M utility function for this agent? Would $\sqrt{u(W)}$ be a valid vN–M utility function for this agent?

4) Suppose an individual has a utility function:

$$u(W) = \ln W$$

The individual faces a gamble where with probability 0.5 their wealth is 100 and with probability 0.5 their wealth is 64.

(a) Describe a general level of coverage X that the individual would purchase as a function of the premium r .

(a) At what level of r would we expect the individual to purchase full insurance coverage $X = 36$?

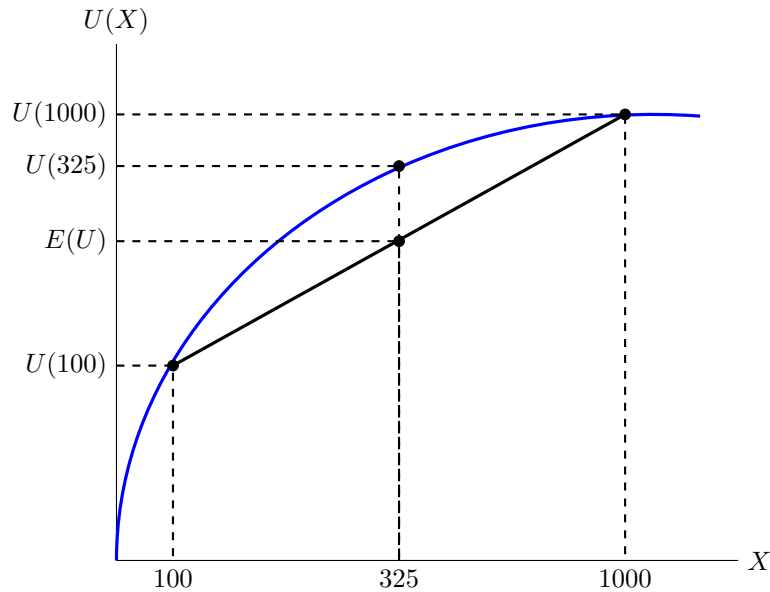
(a) At what level of r would the individual purchase no insurance coverage $X = 0$.

8 Solution

1) (a) This is a fair gamble, since the expected value of the first option is the same as value of the second option:

$$(1/4)(1000) + (3/4)(100) = 325$$

We know that a risk-averse individual (who has some concave vN-M utility function) will reject a fair gamble since the utility of option B will exceed that of option A which is represented as the midpoint of the gamble:



(b) A payment of 320 BD means that its possible for individual to earn more by engaging in the bet since the expected value is higher. Intuitively, this depends on how much risk the person is willing to tolerate.

Graphically, a higher degree of risk aversion implies a greater curvature, which means the CE will be further away from the expected value.

2) For $u(W) = 1/W$, it follows that:

$$\begin{aligned}\rho &= -\frac{u''}{u'} \\ &= -\frac{-\frac{2}{W^3}}{\frac{1}{W^2}} \\ &= \frac{2}{W^3} \cdot W^2 \\ &= \frac{2}{W}\end{aligned}$$

This shows that the individual is risk averse.

3) In expected utility theory, only positive affine transformations of a vN-M utility function preserves preferences over lotteries. That is:

$$a + bu(W), \quad b > 0$$

On the other hand, square rooting is a monotonic transformation, which would disrupt the preference structure of a vN-M utility function.

Consider a basic example where an individual is choosing between two options:

- Option A: 50% chance of $W = 0$ and 50% chance of $W = 9$.
- Option B: 100% chance of $W = 4$

To keep things simple, let the individual's vN-M utility function be defined as:

$$u(W) = W$$

Evaluating the utility from those options:

$$EU(A) = 0.5(0) + 0.5(9) = 4.5$$

$$EU(B) = 4$$

We can clearly see that option A is preferred since $EU(A) > EU(B)$.

Now consider if we transformed the equation monotonically, such as $u(W) = \sqrt{W}$. Evaluating the utility from those options:

$$EU(A) = 0.5(\sqrt{0}) + 0.5(\sqrt{9}) = 1.5$$

$$EU(B) = \sqrt{4} = 2$$

Now we can see that option B is preferred since $EU(B) > EU(A)$.

Since the rankings can flip for a monotonic transformation, then this property violates the assumptions of the vN-M utility function. On the other hand, a positive affine transformation with $b > 0$ preserves the preferences in all cases.

4) It is important to identify that the utility function has a concave property where $u'(W) > 0$ and $u''(W) < 0$, which means the individual is risk averse.

(a) We can make a general statement about the outcomes with respect of X and r :

- Bad Outcome: $W_L = 64 + X - rX$ at probability 0.5.
- Good Outcome: $W_N = 100 - rX$ at probability 0.5.

The expected utility of the individual facing this uncertainty is:

$$E(U) = 0.5 \ln(64 + X - rX) + 0.5 \ln(100 - rX)$$

The individual is maximising their expected utility with respect to their coverage X . The optimal level is determined under first order conditions:

$$\frac{\partial E(U)}{\partial X} = 0.5 \frac{1}{(64 + X - rX)}(1 - r) - 0.5 \frac{1}{(100 - rX)}r = 0$$

$$0.5 \frac{1}{(64 + X - rX)}(1 - r) = 0.5 \frac{1}{(100 - rX)}r$$

$$(100 - rX)(1 - r) = (64 + X - rX)r$$

$$100 - 100r - rX + Xr^2 = 64r + rX - Xr^2$$

$$100 - 100r - 64r = 2rX - 2Xr^2$$

$$100 - 164r = 2rX(1 - r)$$

$$X = \frac{100 - 164r}{2r(1 - r)}$$

This describe a general coverage level X as a function of r .

(b) Remember that the gamble shows that the individual will suffer a loss of 36 BD with probability 0.5 (and so their wealth will be 64 BD). This means that full coverage implies $X = 36$.

We discussed in the chapter that when the insurance companies charge a fair rate, a risk averse individual will purchase full coverage. We saw that this occurs when $r = p = 0.5$ and this can be shown from solution in (a):

$$X = \frac{100 - 164(0.5)}{2(0.5)(1 - (0.5))} = 36$$

(c) The level of r required in order for $X = 0$ can be solved from the equation:

$$0 = \frac{100 - 164r}{2r(1 - r)}$$

$$r = 0.609$$